

Bioequivalence of Single Dose Dapagliflozin, Glimepiride and Metformin Tablets in Healthy Adult Volunteers: Results from a Randomized, Two-Way Crossover, Open-Label Study

Prakadeesh Bharathi¹, Sudershan Kumar², Prachi Ahire¹, Neeraj Markandeywar¹, Amey Mane¹, Arshad Khuroo²

¹Sun Pharma Laboratories Limited, Mumbai, India; ²Sun Pharmaceutical Industries Limited, Haryana, India

INTRODUCTION

- Type 2 diabetes mellitus (T2DM) is a metabolic disorder that is associated with increased morbidity and mortality.¹
- Tight glycaemic control is an important aspect in management of T2DM and may require multiple anti-diabetic agents.²
- If the goal of achieving glycaemic control is not achieved by monotherapy, a combination therapy with different mechanisms of action should be initiated promptly.³
- Both Glimepiride and Dapagliflozin significantly improve glycaemic control and Metformin increases insulin sensitivity.^{4,5}
- Dapagliflozin, Glimepiride and Metformin Hydrochloride has complementary mechanisms of action and its fixed dose combination (FDC) may provide patients with T2DM a simplified treatment regimen, thus reducing pill burden and improving compliance and outcomes.⁶

OBJECTIVE

- To assess comparative bioequivalence of FDC that is a bilayer tablet containing Dapagliflozin, Glimepiride, and Metformin (5mg/2mg/1000mg) (by SunPharma Ltd.), compared to the reference products [Oxramet Tablets (Dapagliflozin 5 mg/Metformin HCl 1000mg IR Tablets; by SunPharma Ltd.) and Amaryl Tablets (Glimepiride 2 mg; by Sanofi India Ltd.)] administered separately.

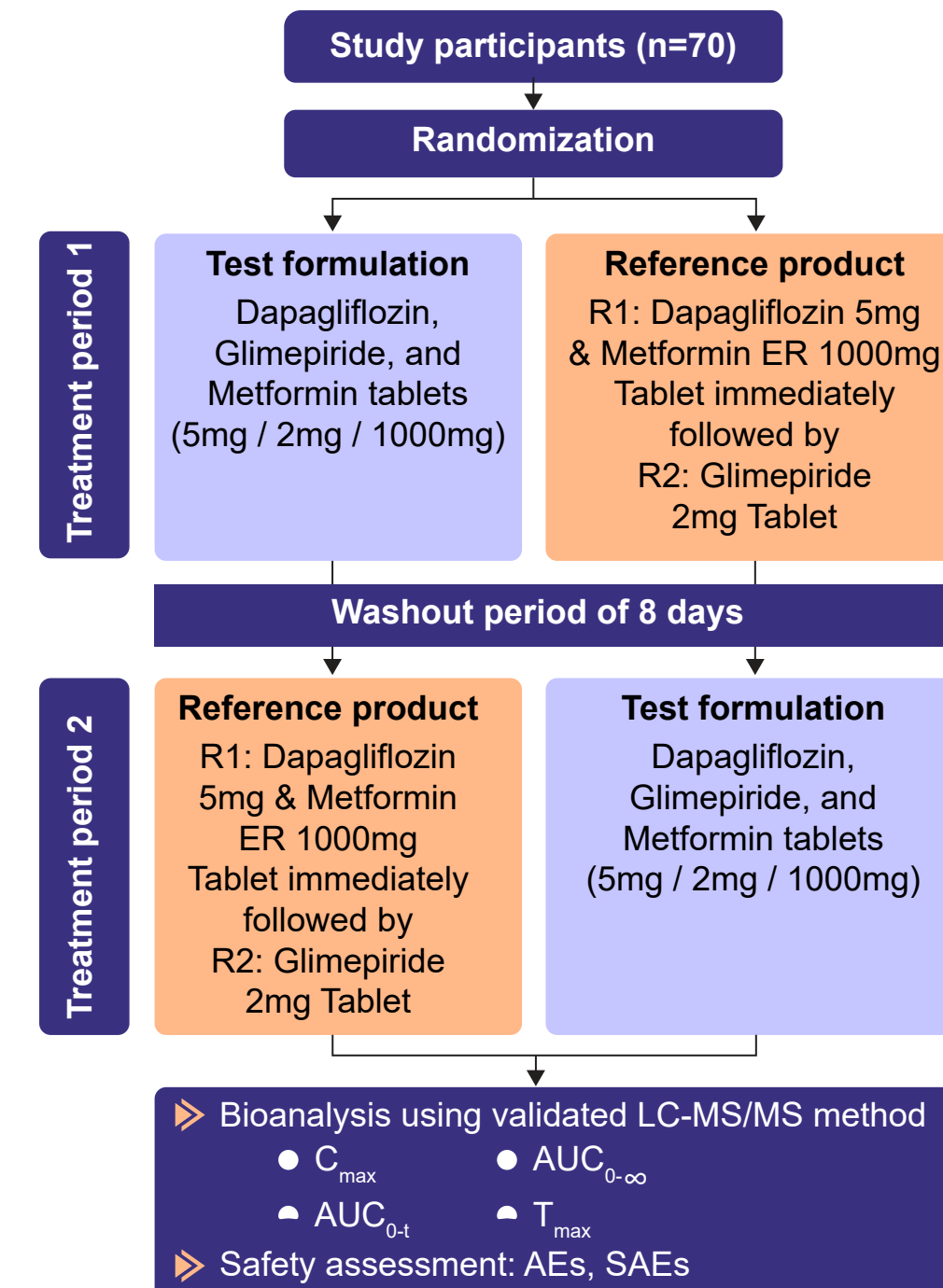
METHODOLOGY

- **Study design:** Open label balanced, randomized, two-treatment, two sequence, two period, single dose, crossover comparative bioavailability study in healthy adult human subjects under fed condition
- **Inclusion Criteria:** Healthy adult humans aged 18-50 years; BMI ≥ 18.50 and < 30.00 kg/m²; Hb ≥ 13.0 g/dL for males or ≥ 12.0 g/dL for females, and who have given written informed consent to participate in this study

Exclusion Criteria:

- Subjects with history of hypersensitivity, recurrent impaired renal function or renal disease or dialysis, any infection, fever, vomiting and/ or diarrhoea, malaise, myalgia, respiratory distress or abdominal pain, drug induced rash and/or pruritus, serious medical illnesses, drug dependence or excessive alcohol intake.
- Subjects having evidence of organ dysfunction or any clinically significant abnormality, had used any medication within 30 days, had donated blood or received an investigational medicinal product, or consumed alcohol, grapefruit juice and or grape fruit supplements 48 hours prior to admission.
- Subjects who were regular smoker, had inability to communicate well, or were pregnant or breast feeding females.

Figure 1: Study design flowchart



RESULTS

- Overall, 57 subjects completed study out of 70 enrolled.

Pharmacokinetic evaluation

Table 1: Summary of pharmacokinetic results of Test and Reference products

Drug	Parameter	Geometric LSM		% Ratio (T/R)	90% CI
		Test product	Reference product		
Dapagliflozin	C _{max} (ng/mL)	45.361	50.188	90.48	85.39 - 95.87
	AUC _{0-t} (hr×ng/mL)	239.8224	239.3555	100.23	98.41 - 102.09
	AUC _{0-∞} (hr×ng/mL)	253.7978	254.4905	99.76	97.81 - 101.74
	*#T _{max} (h)	2.667 (0.667 - 10.000)	3.000 (1.000 - 10.000)	–	–
Glimepiride	C _{max} (ng/mL)	170.532	153.232	111.33	105.18 - 117.85
	AUC _{0-t} (hr×ng/mL)	1071.5063	1061.0563	101.10	96.38 - 106.06
	AUC _{0-∞} (hr×ng/mL)	1009.9524	994.6711	101.42	96.61 - 106.46
	*#T _{max} (h)	3.000 (1.000 - 10.000)	3.667 (1.667 - 10.000)	–	–
Metformin	C _{max} (ng/mL)	1557.82	1544.58	100.86	94.14 - 101.43
	AUC _{0-t} (hr×ng/mL)	12521.4761	12817.8393	97.76	94.18 - 101.47
	AUC _{0-∞} (hr×ng/mL)	1009.9524	994.6711	101.42	96.61 - 106.46
	*#T _{max} (h)	3.667 (1.000 - 7.000)	4.000 (1.333 - 7.000)	–	–

*Arithmetic mean presented; #Median (Min-Max) presented

SAFETY RESULTS

- A total of 11 AEs were recorded, 6 with test product and 5 with reference products.
- All AEs were mild to moderate in severity and resolved without sequelae.
- No SAEs were reported in the study.

DISCUSSION AND CONCLUSION

- Single oral dose of FDC tablet of Dapagliflozin (5mg), Metformin (1000mg), and Glimepiride (2mg) is bioequivalent to individual reference products in healthy adults under fed conditions, with pharmacokinetic parameters (AUC, C_{max}) within the 80-125% bioequivalence range.
- Both test and reference products were well tolerated by the study subjects. There was no reported in the study.
- The study supports the FDC as a viable, effective, and well-tolerated treatment option in diabetes care, especially for patients who require multiple antidiabetic agents.

References:

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Abbreviations:

AEs: Adverse events; AUC: Area under the receiver operating characteristic (ROC) Curve; C_{max}: Maximum concentration; FDC: Fixed dose combination; LC-MS: Liquid chromatography–mass spectrometry; LSM: Least-squares mean; SAEs: Serious adverse events; T2DM: Type 2 diabetes mellitus; T_{max}: Time to peak drug concentration.

Figure 2: Mean plasma concentration–time profiles of a) Dapaglifloxacin b) Glimepiride and c) Metformin in Test and Reference products

